



PROBABILITY THEORY

MTM2592-Gr:2

SEVENTH WEEK

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1. For the traffic accidents happening in a highway, the random variables X represents the healthy situation of people and Y represents whether they wore a seat belt such as

$$X = \begin{cases} 0, & \text{dead} \\ 1, & \text{injured} \\ 2, & \text{healthy} \end{cases} \quad Y = \begin{cases} 1, & \text{wore a seat belt} \\ 0, & \text{did not wear a seat belt} \end{cases}$$

$X \backslash Y$	wore a seat belt	did not wear a seat belt
dead	45	230
injured	120	340
healthy	850	1015

According to the given table,

- find the joint probability function of X and Y ,
- find the marginal distributions $g(x)$ and $h(y)$.

2. Let $f(x, y)$ be a joint probability function of X and Y . If $f(1,1) = 0.4$, $f(1,2) = 0.1$, $f(2,1) = 0.2$ and $f(2,2) = 0.3$, find the conditional probability of X given $Y = 1$.

3. A function is given as

$$f(x, y) = \begin{cases} \frac{1}{21}(2x - y), & x = 1, 2, 3, \quad y = 0, 1, \\ 0, & \text{otherwise} \end{cases}$$

- a) Is it a joint probability function?
- b) Find the marginal distributions X and Y .
- c) Compute $P(X \geq 2 \mid Y = 0)$.
- d) Compute $P(X \leq 2 \mid Y = 2)$.
- e) Compute $P(X > Y)$.
- f) Are X and Y independent?

4. A function is given as

$$f(x, y) = \begin{cases} c(2x + 3y), & x = 0, 1, 2, \quad y = 0, 1, \\ 0, & \text{otherwise} \end{cases},$$

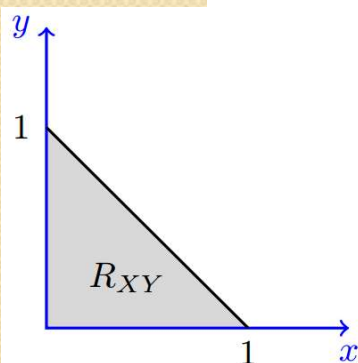
- a) Find c if f a probability function.
- b) Find $P(X)$ and $P(Y)$.
- c) Find $P(Y | X), P(X | Y)$
- d) Compute $P(Y > 0 | X \leq 1)$ and $P(X \leq 1 | Y > 0)$.

5)

Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cx + 1 & x, y \geq 0, x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Show the range of (X, Y) , R_{XY} , in the $x - y$ plane.
2. Find the constant c .
3. Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.
4. Find $P(Y < 2X^2)$.

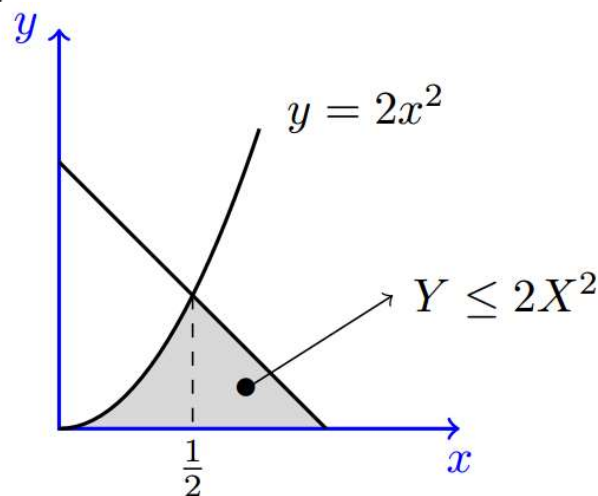


$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y) dx dy \\ &= \int_0^1 \int_0^{1-x} (cx + 1) dy dx \\ &= \int_0^1 (cx + 1)(1 - x) dx \\ &= \frac{1}{2} + \frac{1}{6}c. \end{aligned} \quad \Rightarrow \quad c = 3$$

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^{1-x} (3x+1) dy = (3x+1)(1-x) \quad \Rightarrow g(x) = \begin{cases} (3x+1)(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$h(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^{1-y} (3x+1) dx = \left(\frac{3}{2}(1-y)^2 + (1-y) \right) = \frac{1}{2}(1-y)(5-3y) \quad \Rightarrow h(y) = \begin{cases} \frac{1}{2}(1-y)(5-3y) & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

d) $P(Y < 2X^2) = ?$



$$\begin{aligned}
 P(Y < 2X^2) &= \int_{-\infty}^{\infty} \int_{-\infty}^{2x^2} f_{XY}(x, y) dy dx \\
 &= \int_0^1 \int_0^{\min(2x^2, 1-x)} (3x+1) dy dx \\
 &= \int_0^1 (3x+1) \min(2x^2, 1-x) dx \\
 &= \int_0^{\frac{1}{2}} 2x^2(3x+1) dx + \int_{\frac{1}{2}}^1 (3x+1)(1-x) dx \\
 &= \frac{53}{96}.
 \end{aligned}$$

6. The joint probability distribution of X and Y random variables is given in table.

	1	2	3	4	Sum
-1	0	0	0.1	0.2	0.3
0	0.1	0.1	0	0.1	0.3
1	0.2	0	0.1	0.1	0.4
Sum	0.3	0.1	0.2	0.4	1

- a) Find the marginal distributions X and Y .
- b) Find $E(X)$ and $E(Y)$,
- c) Find $E(XY)$,
- d) Are X and Y independent?
- e) Find $E(3X+1)$, $E(2X+4Y)$, $E(X^2+Y)$;
- f) Find $E(Y | X = 0)$,
- g) Find $Var(Y | X = 0)$.

7) Marginal distribution of X random variable, conditional distribution of Y given X=0 and conditional distribution of Y given X=2 are given by.

$$f(x) = \begin{cases} 0.4, & x = 0 \\ 0.6, & x = 2 \\ 0, & \text{Elsewhere} \end{cases}$$

$$f(Y | X = 0) = \begin{cases} 0.8, & y = 0 \\ 0.2, & y = 1 \\ 0, & \text{Elsewhere} \end{cases}$$

$$f(Y | X = 2) = \begin{cases} 0.5, & y = 0 \\ 0.5, & y = 1 \\ 0, & \text{Elsewhere} \end{cases}$$

- a) Find the joint probability distribution of X and Y.
- b) Find $E(Y | X = 2)$,
- c) Find $f(X | Y = 0)$,
- d) Find $\text{Var}(X | Y = 0)$,
- e) Find $U = E(X | Y)$,
- f) Find $E(U) = E[E(X | Y)]$.

8) If the joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{3}(x + y), & 0 \leq x \leq 1, \ 0 \leq y \leq 2, \\ 0 & \text{elsewhere} \end{cases}$$

Find the correlation coefficient ($\text{Corr}(X, Y)$) of X and Y.



9) If the random variables X, Y and Z have the means $E(X)=-2$, $E(Y)=1$, $E(Z)=3$, the variances $\text{Var}(X)=1$, $\text{Var}(Y)=1$, $\text{Var}(Z)=4$, and the covariances $\text{Cov}(X,Y)=-0.05$, $\text{Cov}(X,Z)=0.10$ and $\text{Cov}(Y,Z)=1$.

a) Find the mean and the variance of $U_1=X-2Y-3Z$

b) If $U_2=5X-2Y$, find the covariance of $\text{Cov}(U_1, U_2)$.

Ex: Let X,Y be jointly continuous random variables with joint PDF:

$$f(x,y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- Are X and Y independent?
- Find conditional distribution of X given $Y = 0,5$.
- Find conditional distribution of Y given $X = 2$.
- Find $P(X > 2 | Y > 0,5)$.
- Find $P(Y < 0,4 | X > 2)$.
- Find the mean of X and Y random variables, and use these values to determine the dependency.
- Find $\text{Cov}(X,Y)$.
- Find correlation coefficient.
- Find conditional mean and variance of X given $Y=0.5$.
- Find conditional mean and variance of Y given $X=2$.

Solution:

a)

$$f(x,y) \stackrel{?}{=} g(x) \cdot h(y) \Rightarrow \frac{3}{23}(x^2 - y) \neq \frac{3}{46}(2x^2 - 1) \cdot \frac{1}{23}(26 - 6y)$$

$$g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$



X and Y random variables are dependent.

$$f(x,y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

b) Find conditional distribution of X given $Y = 0,5$.

$$P(X = x | Y = 0,5) = f(x|0,5) = \frac{f(x, 0,5)}{h(0,5)} = \frac{\frac{3}{23}\left(x^2 - \frac{1}{2}\right)}{\frac{1}{23}(26 - 6 \cdot (0,5))} = \frac{3}{46}(2x^2 - 1) \Rightarrow f(x|0,5) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

c) Find conditional distribution of Y given $X = 2$.

$$P(Y = y | X = 2) = f(y|2) = \frac{f(2, y)}{g(2)} = \frac{\frac{3}{23}(4 - y)}{\frac{3}{46}(8 - 1)} = \frac{2}{7}(4 - y) \Rightarrow f(y|2) = \begin{cases} \frac{2}{7}(4 - y) & 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

d) Find $P(X > 2|Y > 0,5)$.

$$P(X > 2|Y > 0.5) = \frac{P(X > 2, Y > 0.5)}{P(Y > 0.5)} = \frac{\int_2^3 \int_{0.5}^1 \frac{3}{23} (x^2 - y) dy dx}{\int_{0.5}^1 \frac{1}{23} (26 - 6y) dy}$$

$$\Rightarrow P(X > 2|Y > 0.5) = 0,364$$

e) Find $P(Y < 0,4|X > 2)$

$$P(Y < 0.4|X > 2) = \frac{P(X > 2, Y < 0.4)}{P(X > 2)} = \frac{\int_2^3 \int_0^{0.4} \frac{3}{23} (x^2 - y) dy dx}{\int_2^3 \frac{3}{46} (2x^2 - 1) dx}$$

$$P(Y < 0.4|X > 2) = \frac{\frac{3}{23} \int_2^3 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^{0.4} dx}{\frac{3}{46} \left| \frac{2x^3}{3} - x \right|_2^3} = \frac{\frac{3}{23} \int_2^3 (0.4x^2 - 0.08) dx}{\frac{35}{46}} = \frac{0.32}{0.76} = 0.42$$

$$f(x,y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

f) Find the mean of X and Y random variables, and use these values to determine the dependency.

$$E(X) = \mu_x = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \cdot f(x,y) dy dx = \int_{-\infty}^{\infty} x \cdot g(x) dx = \int_1^3 x \cdot \frac{3}{46}(2x^2 - 1) dx = \frac{48}{23}$$

$$E(Y) = \mu_y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \cdot f(x,y) dy dx = \int_{-\infty}^{\infty} y \cdot h(y) dy = \int_0^1 y \cdot \frac{1}{23}(26 - 6y) dy = \frac{11}{23}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f(x,y) dy dx = \int_1^3 \int_0^1 xy \cdot \frac{3}{23}(x^2 - y) dy dx = \frac{26}{23}$$

$$\left. \begin{array}{l} E(XY) \neq E(X)E(Y) \\ \frac{26}{23} \neq \frac{48}{23} \cdot \frac{11}{23} \end{array} \right\} \begin{array}{l} X \text{ and } Y \text{ are dependent} \end{array}$$

g) Find Cov(X,Y).

$$\sigma_{xy} = \text{Cov}(X,Y) = E(XY) - \mu_x \mu_y = \frac{26}{23} - \frac{48}{23} \cdot \frac{11}{23} = \frac{70}{529} = 0,1323$$



There is a weak positive relationship between X and Y.

$$f(x,y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

h) Find correlation coefficient.

$$\rho_{xy} = \frac{Cov(X,Y)}{\sqrt{Var(X) \cdot Var(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \quad Var(X) = E(X^2) - \mu_x^2 \quad Var(Y) = E(Y^2) - \mu_y^2 \quad \mu_x = \frac{60}{23} \quad \mu_y = \frac{11}{23}$$

$$E(X^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 \cdot f(x,y) dy dx = \int_{-\infty}^{\infty} x^2 \cdot g(x) dx = \int_1^3 x^2 \cdot \frac{3}{46}(2x^2 - 1) dx = \frac{661}{115}$$

$$Var(X) = \frac{661}{115} - \left(\frac{48}{23}\right)^2 = 1,3924$$

$$E(Y^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 \cdot f(x,y) dy dx = \int_{-\infty}^{\infty} y^2 \cdot h(y) dy = \int_0^1 y^2 \cdot \frac{1}{23}(26 - 6y) dy = \frac{43}{138}$$

$$Var(Y) = \frac{43}{138} - \left(\frac{11}{23}\right)^2 = 0,0829$$

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{0,1323}{\sqrt{(1,3924)(0,0829)}} = 0,3894$$

- i) Find conditional mean and variance of X given Y=0.5.
 j) Find conditional mean and variance of Y given X=2.

$$i) \quad f(x|0,5) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

$$\mu_{x|0,5} = E(X|0,5) = \int_{-\infty}^{\infty} x \cdot f(x|0,5) dx = \int_1^3 x \cdot \frac{3}{46}(2x^2 - 1) dx = \frac{48}{23}$$

$$E(X^2|0,5) = \int_{-\infty}^{\infty} x^2 \cdot f(x|0,5) dx = \int_1^3 x^2 \cdot \frac{3}{46}(2x^2 - 1) dx = \frac{661}{115}$$

$$\sigma_{x|0,5}^2 = E(X^2|0,5) - (\mu_{x|0,5})^2 = \frac{661}{115} - \left(\frac{48}{23}\right)^2 = \frac{3683}{2645} = 1,3924$$

$$j) \quad f(y|2) = \begin{cases} \frac{2}{7}(4-y) & 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\mu_{y|2} = E(Y|2) = \int_{-\infty}^{\infty} y \cdot f(y|2) dy = \int_0^1 y \cdot \frac{2}{7}(4-y) dy = \frac{3}{14}$$

$$E(Y^2|2) = \int_{-\infty}^{\infty} y^2 \cdot f(y|2) dy = \int_0^1 y^2 \cdot \frac{2}{7}(4-y) dy = \frac{13}{42}$$

$$\sigma_{y|2}^2 = E(Y^2|2) - (\mu_{y|2})^2 = \frac{13}{42} - \left(\frac{3}{14}\right)^2 = \frac{155}{588} = 0,2636$$

Örnek: 3 mavi, 2 kırmızı ve 3 yeşil kalemin olduğu bir kutudan iki kalem, seçilecektir. Eğer X rasgele değişkeni seçilen mavi kalem sayısını, Y ise seçilen kırmızı kalem sayısını gösterirse:

- $f(x, y)$ ortak olasılık fonksiyonunu tanımlayınız
- $A = \{(x, y) | x + y \leq 1\}$ olmak üzere $P((X, Y) \in A)$ olasılığını bulunuz.
- Ortak birikimli olasılık fonksiyonunu belirleyiniz.
- X ve Y rastgele değişkenlerinin marjinal olasılık fonksiyonlarını belirleyiniz.

Çözüm:

$$a) \quad P(X=x, Y=y) = f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-(x+y)}}{\binom{8}{2}} \quad \begin{matrix} x=0,1,2 \\ y=0,1,2 \end{matrix}$$

		X		
		0	1	2
Y	0	3/28	9/28	3/28
	1	6/28	6/28	0
	2	1/28	0	0

b)

$$P((X,Y) \in A) = f(0,0) + f(0,1) + f(1,0) = \frac{3}{28} + \frac{6}{28} + \frac{9}{28} = \frac{9}{14} = 0,643$$

c)

$$F(x,y) = \begin{cases} 0 & x < 0, y < 0 \\ 3/28 & 0 \leq x < 1, 0 \leq y < 1 \\ 9/28 & 0 \leq x < 1, 1 \leq y < 2 \\ 10/28 & 0 \leq x < 1, y \geq 2 \\ 12/28 & 1 \leq x < 2, 0 \leq y < 1 \\ 24/28 & 1 \leq x < 2, 1 \leq y < 2 \\ 25/28 & 1 \leq x < 2, y \geq 2 \\ 15/28 & x \geq 2, 0 \leq y < 1 \\ 27/28 & x \geq 2, 1 \leq y < 2 \\ 1 & x \geq 2, y \geq 2 \end{cases}$$

$$F(1,1) = \frac{24}{28}, F(1,2) = \frac{25}{28}, F(0,2) = \frac{10}{28}$$

	X		
	0	1	2
Y	0	3/28	9/28
	1	6/28	6/28
	2	1/28	0

$$d) \quad g(x) = \sum_{y=0}^2 f(x, y) = f(x, 0) + f(x, 1) + f(x, 2),$$

$$X=0 \Rightarrow g(0) = \sum_{y=0}^2 f(0, y) = f(0, 0) + f(0, 1) + f(0, 2) = \frac{10}{28}$$

$$X=1 \Rightarrow g(1) = \sum_{y=0}^2 f(1, y) = f(1, 0) + f(1, 1) + f(1, 2) = \frac{15}{28}$$

$$X=2 \Rightarrow g(2) = \sum_{y=0}^2 f(2, y) = f(2, 0) + f(2, 1) + f(2, 2) = \frac{3}{28}$$

$$h(y) = \sum_{x=0}^2 f(x, y) = f(0, y) + f(1, y) + f(2, y)$$

$$Y=0 \Rightarrow h(0) = \sum_{x=0}^2 f(x, 0) = f(0, 0) + f(1, 0) + f(2, 0) = \frac{15}{28}$$

$$Y=1 \Rightarrow h(1) = \sum_{x=0}^2 f(x, 1) = f(0, 1) + f(1, 1) + f(2, 1) = \frac{12}{28}$$

$$Y=2 \Rightarrow h(2) = \sum_{x=0}^2 f(x, 2) = f(0, 2) + f(1, 2) + f(2, 2) = \frac{1}{28}$$

X	0	1	2
g(x)	10/28	15/28	3/28

Y	0	1	2
h(y)	15/28	12/28	1/28

		X			h(y)
		0	1	2	
Y	0	3/28	9/28	3/28	15/28
	1	6/28	6/28	0	12/28
	2	1/28	0	0	1/28
g(x)		10/28	15/28	3/28	

Örnek: X,Y rastgele değişkeninin ortak olasılık yoğunluk fonksiyonu şöyle verilmiştir.

$$f(x, y) = \begin{cases} k(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{diger} \end{cases}$$

- a) k ne olmalıdır?
- b) Ortak birikimli olasılık fonksiyonunu belirleyiniz.
- c) X ve Y rastgele değişkenlerinin marjinal olasılık fonksiyonlarını belirleyiniz.
- d) $P(X < 2, Y < 0.5)$ ve $P[(X+Y) < 2]$ olasılıklarını hesaplayınız

Çözüm:

a) $i) f(x, y) \geq 0 \quad ii) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_0^1 \int_1^3 k(x^2 - y) dy dx = k \int_1^3 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^1 dx = k \int_1^3 \left(x^2 - \frac{1}{2} \right) dx = k \left[\frac{x^3}{3} - \frac{x}{2} \right]_1^3 = 1$$

$$\Rightarrow k \left[\frac{27}{3} - \frac{3}{2} - \left[\frac{1}{3} - \frac{1}{2} \right] \right] = k \left[\frac{23}{3} \right] = 1 \Rightarrow k = \frac{3}{23}$$

$$f(x, y) = \begin{cases} \frac{3}{23}(x^2 - y) & 1 < x < 3 \quad 0 < y < 1 \\ 0 & \text{diger} \end{cases}$$

$$b) \quad F(x, y) = \begin{cases} 0 & x \leq 1, y \leq 0 \\ F_1(x, y) & 1 < x < 3, 0 < y < 1 \\ F_2(x) & 1 < x < 3, y \geq 1 \\ F_3(y) & x \geq 3, 0 < y < 1 \\ 1 & x \geq 3, y \geq 1 \end{cases}$$

$$F_1(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^x \int_0^y \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^x (u^2 v - \frac{v^2}{2}) \Big|_0^y du$$

$$= \frac{3}{23} \int_1^x (u^2 y - \frac{y^2}{2}) du = \frac{3}{23} \left[\frac{u^3 y}{3} - \frac{uy^2}{2} \right]_1^x = \frac{3}{23} \left[\frac{x^3 y}{3} - \frac{xy^2}{2} - \left(\frac{y}{3} - \frac{y^2}{2} \right) \right] = \frac{2x^3 y - 3xy^2 + 3y^2 - 2y}{46}$$

$$F_2(x) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^x \int_1^1 \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^x (u^2 v - \frac{v^2}{2}) \Big|_0^y du$$

$$= \frac{3}{23} \int_1^x (u^2 - \frac{1}{2}) du = \frac{3}{23} \left[\frac{u^3}{3} - \frac{u}{2} \right]_1^x = \frac{3}{23} \left[\frac{x^3}{3} - \frac{x}{2} - \left(\frac{1}{3} - \frac{1}{2} \right) \right] = \frac{2x^3 - 3x + 1}{46}$$

$$F_3(y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx = \int_1^3 \int_0^y \frac{3}{23} (u^2 - v) dv du = \frac{3}{23} \int_1^3 (u^2 v - \frac{v^2}{2}) \Big|_0^y du$$

$$= \frac{3}{23} \int_1^3 (u^2 y - \frac{y^2}{2}) du = \frac{3}{23} \left[\frac{u^3 y}{3} - \frac{uy^2}{2} \right]_1^3 = \frac{3}{23} \left[\frac{27y}{3} - \frac{3y^2}{2} - \left(\frac{y}{3} - \frac{y^2}{2} \right) \right] = \frac{52y - 6y^2}{46}$$

$$F(x, y) = \begin{cases} 0 & x \leq 1, y \leq 0 \\ \frac{2x^3y - 3xy^2 + 3y^2 - 2y}{46} & 1 < x < 3, 0 < y < 1 \\ \frac{2x^3 - 3x + 1}{46} & 1 < x < 3, y \geq 1 \\ \frac{52y - 6y^2}{46} & x \geq 3, 0 < y < 1 \\ 1 & x \geq 3, y \geq 1 \end{cases}$$

c)

$$g(x) = \int_0^1 \frac{3}{23}(x^2 - y)dy = \frac{3}{23} \left[x^2y - \frac{y^2}{2} \right]_0^1 = \frac{3}{46}(2x^2 - 1)$$

$$\Rightarrow g(x) = \begin{cases} \frac{3}{46}(2x^2 - 1) & 1 \leq x \leq 3 \\ 0 & \text{diger} \end{cases}$$

$$h(y) = \int_1^3 \frac{3}{23}(x^2 - y)dx = \frac{3}{23} \left[\frac{x^3}{3} - xy \right]_1^3 = \frac{3}{23} \left[\frac{27}{3} - 3y - \left(\frac{1}{3} - y \right) \right]$$

$$= \frac{3}{23} \left[\frac{27 - 9y - 1 + 3y}{3} \right] = \frac{1}{23} [26 - 6y]$$

$$\Rightarrow h(y) = \begin{cases} \frac{1}{23}(26 - 6y) & 0 \leq y \leq 1 \\ 0 & \text{diger} \end{cases}$$

d)

$$P(X < 2, Y < 0.5) = \int_1^2 \int_0^{0.5} \frac{3}{23}(x^2 - y) \, dy dx = \frac{3}{23} \int_1^2 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^{0.5} dx$$

$$P(X < 2, Y < 0.5) = \frac{3}{23} \int_1^2 (0.5x^2 - 0.125) dx = \frac{3}{23} \left(\frac{0.5x^3}{3} - 0.125x \right) \Big|_1^2$$

$$P(X < 2, Y < 0.5) = 0,1685$$

$$P[(X + Y) < 2] = \int_1^2 \int_0^{2-x} \frac{3}{23}(x^2 - y) \, dy dx = \frac{3}{23} \int_1^2 \left(x^2 y - \frac{y^2}{2} \right) \Big|_0^{2-x} dx$$

$$\frac{3}{23} \int_1^2 \left[x^2(2-x) - \frac{(2-x)^2}{2} \right] dx = \frac{3}{23} \int_1^2 \left(2x^2 - x^3 - \left(\frac{4-4x+x^2}{2} \right) \right) dx$$

$$P[(X + Y) < 2] = \frac{3}{23} \left[\frac{2x^3}{3} - \frac{x^4}{4} - \frac{4x}{2} + x^2 - \frac{x^3}{6} \right] \Big|_1^2 \Rightarrow P[(X + Y) < 2] = 0,75$$

Örnek: 3 mavi, 2 kırmızı ve 3 yeşil kalemin olduğu bir kutudan iki kalem, seçilecektir. Eğer X rasgele değişkeni seçilen mavi kalem sayısını, Y ise kırmızı kalem sayısını gösterirse,

- 1 mavi kalem seçildiği bilindiğinde seçilecek kırmızı kalem sayısının koşullu dağılımını bulunuz.
- Hiç kırmızı kalem seçilmediği bilindiğinde seçilecek mavi kalem sayısının koşullu dağılımını bulunuz.
- X ve Y rastgele değişkenleri bağımsız mıdır?
- X ve Y rastgele değişkenlerinin ortalamalarını hesaplayınız, bunlardan faydalananarak bağımsızlığa bakınız.
- Cov(X,Y) yi hesaplayıp yorumlayınız.
- Korelasyon katsayısını hesaplayıp yorumlayınız.

Çözüm:

a)

		X			h(y)
		0	1	2	
Y	0	3/28	9/28	3/28	15/28
	1	6/28	6/28	0	12/28
	2	1/28	0	0	1/28
g(x)		10/28	15/28	3/28	

$$P(Y = y|X = 1) = f(y|1) = \frac{f(1,y)}{g(1)} = \frac{f(1,y)}{15/28} = \frac{28}{15} f(1,y)$$

$$Y = 0 \Rightarrow f(0|1) = \frac{28}{12} f(1,0) = \frac{28}{15} \frac{9}{28} = \frac{9}{15} = \frac{3}{5}$$

$$Y = 1 \Rightarrow f(1|1) = \frac{28}{12} f(1,1) = \frac{28}{15} \frac{6}{28} = \frac{6}{15} = \frac{2}{5}$$

$$Y = 2 \Rightarrow f(2|1) = \frac{28}{15} f(1,2) = \frac{28}{15} 0 = 0$$

Y	0	1
$f(y 1)$	3/5	2/5

b) Hiç kırmızı kalem seçilmediği bilindiğinde seçilecek mavi kalem sayısının koşullu dağılımını bulunuz.

$$P(X=x|Y=0) = f(x|0) = \frac{f(x,0)}{h(0)} = \frac{f(x,0)}{15/28} = \frac{28}{15} f(x,0)$$

$$X=0 \Rightarrow f(0|0) = \frac{28}{15} f(0,0) = \frac{28}{15} \cdot \frac{3}{28} = \frac{1}{5}$$

$$X=1 \Rightarrow f(1|0) = \frac{28}{15} f(1,0) = \frac{28}{15} \cdot \frac{9}{28} = \frac{3}{5}$$

$$X=2 \Rightarrow f(2|0) = \frac{28}{15} f(2,0) = \frac{28}{15} \cdot \frac{3}{28} = \frac{1}{5}$$

X	0	1	2
$f(x 0)$	1/5	3/5	1/5

		X			h(y)
Y		0	1	2	
	0	3/28	9/28	3/28	15/28
	1	6/28	6/28	0	12/28
	2	1/28	0	0	1/28
g(x)		10/28	15/28	3/28	

c) X ve Y rastgele değişkenleri bağımsız mıdır?

$\forall (x, y)$ için $f(x, y) = g(x) \cdot h(y)$ olmalıdır.

$(x, y) = (0, 0)$ için $f(0, 0) = g(0) \cdot h(0)$ mi?

$$\frac{3}{28} \neq \frac{15}{28} \cdot \frac{10}{28}$$



X ve Y rastgele değişkenleri bağımlıdır.

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d) X ve Y rastgele değişkenlerinin ortalamalarını hesaplayınız, bunlardan faydalanarak bağımsızlığa bakınız.

$$E(X) = \mu_x = \sum_{x=0}^2 \sum_{y=0}^2 x \cdot f(x, y) = \sum_{x=0}^2 x \left(\sum_{y=0}^2 f(x, y) \right)$$

$$= \sum_{x=0}^2 x \cdot g(x) = 0 \cdot \frac{10}{28} + 1 \cdot \frac{15}{28} + 2 \cdot \frac{3}{28} = \frac{21}{28} = \frac{3}{4}$$

$$E(Y) = \mu_y = \sum_{x=0}^2 \sum_{y=0}^2 y \cdot f(x, y) = \sum_{y=0}^2 y \left(\sum_{x=0}^2 f(x, y) \right)$$

$$= \sum_{y=0}^2 y \cdot h(y) = 0 \cdot \frac{15}{28} + 1 \cdot \frac{12}{28} + 2 \cdot \frac{1}{28} = \frac{14}{28} = \frac{1}{2}$$

$$E(XY) \stackrel{?}{=} E(X) \cdot E(Y)$$

$$E(XY) = \sum_{x=0}^2 \sum_{y=0}^2 x \cdot y \cdot f(x, y) = \sum_{x=0}^2 x \left(\sum_{y=0}^2 y \cdot f(x, y) \right) = \sum_{x=0}^2 x \cdot (0 \cdot f(x, 0) + 1 \cdot f(x, 1) + 2 \cdot f(x, 2))$$

$$= 0 \cdot (0 + 1 \cdot f(0, 1) + 2 \cdot f(0, 2)) + 1 \cdot (0 + 1 \cdot f(1, 1) + 2 \cdot f(1, 2)) + 2 \cdot (0 + 1 \cdot f(2, 1) + 2 \cdot f(2, 2))$$

$$\Rightarrow E(XY) = \left(\frac{6}{28} + 2 \cdot 0 \right) + 2 \cdot (0 + 2 \cdot 0) = \frac{6}{28}$$

$$\frac{6}{28} \neq \frac{3}{4} \cdot \frac{1}{2} \text{ olduğundan X ve Y bağımlıdır.}$$

		X			h(y)
		0	1	2	
Y	0	3/28	9/28	3/28	15/28
	1	6/28	6/28	0	12/28
	2	1/28	0	0	1/28
g(x)		10/28	15/28	3/28	

e) Cov(X,Y) yi hesaplayıp yorumlayınız.

$$E(XY) = \frac{6}{28}$$

$$\sigma_{xy} = Cov(X,Y) = E(XY) - \mu_x \mu_y = \frac{6}{28} - \frac{3}{4} \cdot \frac{1}{2} = -\frac{9}{56} = -0,1607$$



X ile Y arasında negatif ilişki vardır.

f) Korelasyon katsayısını hesaplayıp yorumlayınız.

$$\rho_{xy} = \frac{Cov(X,Y)}{\sqrt{Var(X) \cdot Var(Y)}} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \quad E(X) = \mu_x = \frac{3}{4} \quad E(Y) = \mu_y = \frac{1}{2}$$

$$Var(X) = E(X^2) - \mu_x^2$$

$$E(X^2) = \sum_{x=0}^2 \sum_{y=0}^2 x^2 \cdot f(x,y) = \sum_{x=0}^2 x^2 \left(\sum_{y=0}^2 f(x,y) \right) = \sum_{x=0}^2 x^2 \cdot g(x) = 0^2 \cdot \frac{10}{28} + 1^2 \cdot \frac{15}{28} + 2^2 \cdot \frac{3}{28} = \frac{27}{28}$$

$$E(Y^2) = \sum_{x=0}^2 \sum_{y=0}^2 y^2 \cdot f(x,y) = \sum_{y=0}^2 y^2 \left(\sum_{x=0}^2 f(x,y) \right) = \sum_{y=0}^2 y^2 \cdot h(y) = 0^2 \cdot \frac{15}{28} + 1^2 \cdot \frac{12}{28} + 2^2 \cdot \frac{1}{28} = \frac{16}{28}$$

$$Var(X) = E(X^2) - \mu_x^2$$

$$= \frac{27}{28} - \left(\frac{3}{4}\right)^2 = \frac{45}{112}$$

$$Var(Y) = E(Y^2) - \mu_y^2$$

$$= \frac{16}{28} - \left(\frac{1}{2}\right)^2 = \frac{9}{28}$$



$$\rho_{xy} = \frac{-18}{\sqrt{45 \cdot 36}} = \frac{-\sqrt{5}}{5} = -0,4472$$

		X			h(y)
		0	1	2	
Y	0	3/28	9/28	3/28	15/28
	1	6/28	6/28	0	12/28
	2	1/28	0	0	1/28
g(x)		10/28	15/28	3/28	